NETWORK INFRASTRUCTURE REPORT

Summit Care Medical Clinic Network Architecture Master Blueprint

This document presents a complete network modernization blueprint for Summit Care Medical Clinic, outlining the strategy, architecture, deployment planning, security controls, documentation standards, and recovery design used to transform a small office healthcare environment into a more structured enterprise-grade infrastructure. Summit Care Medical Clinic is used here as a fictionalized organization so the full workflow can be presented clearly while keeping actual production identities and sensitive organizational details private.

Enterprise Network Upgrade

Hybrid Cloud, Security & Recovery

By Sidiq Daniel

Project Overview

A full-length technical report covering the planning, research, build standards, implementation approach, and operational structure behind the Summit Care network upgrade.

Report Focus

Focused on architecture, deployment, hybrid identity, segmentation, cybersecurity, documentation, and disaster recovery across one unified healthcare infrastructure workflow.

Table Of Contents

1. Executive Summary

2. Introduction & Project Intent

3. Company Objectives & Gap Analysis

4. Research & Development

5. Assets, Inventory & Procurement

6. Documentation, Diagrams & Topologies

7. Infrastructure Deployment & Configurations

8. Hybrid Cloud Infrastructure & Scalability

9. Network Segmentation, VLANs & Subnetting

10. Cybersecurity, Data Protection & HIPAA Compliance

11. Disaster Recovery & Backup Plan

12. Conclusion

EXECUTIVE SUMMARY

Project Summary For Reviewers

This report documents a fictionalized healthcare infrastructure upgrade used to demonstrate real implementation workflow, engineering thought process, and operational design maturity. The scenario centers on transforming a small office medical environment into a more production-ready enterprise network through structured planning, standardization, hybrid identity, segmented networking, stronger monitoring, and layered recovery design.

What This Project Demonstrates

End-to-end infrastructure thinking across research, procurement, documentation, deployment, Microsoft hybrid identity, network segmentation, cybersecurity, and disaster recovery. It also demonstrates the ability to present technical work in a sanitized, portfolio-safe format for professional review.

Bottom line: this is a full operational blueprint showing how architecture, deployment, admin standards, security controls, and recovery planning were organized into one coherent delivery model.

10

Core technical domains covered

3-2-1

Backup model used for recovery planning

10.7.0.0/16

Private address space used in the segmentation model

Hybrid

Identity model combining local AD and Microsoft Entra

SECTION 1

Introduction & Project Intent

Summit Care Medical Clinic is presented here as a mock healthcare organization used to demonstrate a real-world infrastructure upgrade workflow in a presentation-safe format. The technical approach, engineering logic, deployment sequence, documentation methods, and support standards in this report reflect practical infrastructure work, but the organization name, identifiers, and contextual framing have been intentionally fictionalized so the report can be shared publicly for educational, portfolio, and professional review purposes.

The core challenge addressed in this blueprint is straightforward: take a small office medical environment with limited standardization, uneven security controls, minimal centralized management, and basic network architecture, then redesign it into a more resilient, supportable, segmented, and enterprise-style production environment. The report does not serve as a generic theory document. It is a workflow-oriented design record that explains what was planned, why it was selected, and how the environment was intended to operate once upgraded.

Positioning statement: this master report is best understood as a technical blueprint and implementation narrative. It preserves the substance of real infrastructure work while replacing client-specific naming and production identifiers with a fictionalized healthcare scenario that can be reviewed safely.

SECTION 2

Company Objectives & Gap Analysis

Summit Care Medical Clinic is framed as a New York City healthcare office that handles patient information, office operations, scheduling activity, billing processes, remote collaboration needs, and internal administrative workloads in a HIPAA-aware environment.

Current-State Conditions

The starting environment reflects a small office network built around a basic Verizon router, shared-password wireless access, manually configured desktops, limited endpoint uniformity, minimal centralized controls, and fragmented visibility into users, systems, and recovery readiness.

Target-State Direction

The target environment introduces Windows domain services, identity-based wireless access, stronger segmentation, hybrid Microsoft identity, standardized endpoints, recovery planning, role-based access, cleaner documentation, and a much more supportable operational baseline.

Primary Objectives

- Replace flat small-office networking with segmented enterprise-style architecture.
- Standardize user desktops and laptops for easier provisioning and scaling.
- Introduce Active Directory, DNS, DHCP, NPS/RADIUS, and centralized Windows identity.
- Replace shared wireless passwords with WPA2-Enterprise / identity-backed wireless access.
- Shift from Google Workspace direction to Microsoft 365 and Entra-connected services.
- Support secure remote work, remote support, VPN access, Teams collaboration, and future office growth.

- Improve documentation, asset visibility, backup readiness, and HIPAA-aware administration.

Gap Analysis Summary

Area	Current State	Target State
Identity	Local/basic account handling with inconsistent endpoint management	Windows domain, role-based access, Entra sync, SSO, MFA
Wireless	Shared-password style access	RADIUS-backed enterprise wireless with separated guest access
Infrastructure	Limited central server role maturity	Virtualized server platform with documented roles and baselines
Security	Minimal segmentation and weaker visibility	VLANs, ACLs, firewalling, monitoring, DLP, logging, SSH standards
Recovery	Insufficient layered backup planning	3-2-1 backup strategy, config exports, UPS protection, NAS/cloud copies

SECTION 3

Research & Development

The research phase evaluated the stack before procurement and deployment began. This included hardware platforms, cloud ecosystems, virtualization options, software choices, licensing paths, VPN models, support tooling, and staff readiness. The point was not to collect abstract ideas, but to make practical choices that matched the clinic's needs, budget expectations, and support realities.

Network Platform Research

Cisco and Ubiquiti were compared for routing, switching, controller management, compatibility, support depth, training overhead, and long-term scalability. Cisco offered stronger enterprise networking depth, while Ubiquiti remained useful in the wireless layer due to practical availability and compatibility with the switching plan.

Cloud & Productivity Research

Google Workspace and Microsoft were compared for identity, productivity, email, licensing, security features, and hybrid integration. Microsoft was selected because it aligned more naturally with the Windows domain model, Entra synchronization, Teams, OneDrive, and role-based cloud administration.

Virtualization Research

Proxmox, Hyper-V, and VMware were reviewed for cost, usability, familiarity, virtualization capabilities, and recovery considerations. Proxmox emerged as the practical fit because it was cost-effective, Linux-based, and aligned with existing technical skill sets.

Licensing, Software, and Operational Comparison

Category	Compared Items	Selected Direction
Hypervisors	Proxmox, Hyper-V, VMware	Proxmox for cost, familiarity, and flexible virtualization
Endpoint Deployment	Clonezilla, Rufus, WDS, PXE workflows	Windows ISO + Rufus + standardized USB deployment path
Remote Support	TeamViewer, AnyDesk	Support software selected based on reliability, security, and admin efficiency
Cloud Productivity	Google Workspace, Microsoft 365	Microsoft 365 with Entra P1, Teams, OneDrive, and hybrid identity
Healthcare Fit	General admin tools and cloud software	Tools reviewed for HIPAA-aware operational suitability before adoption

Estimated Cost Considerations

Item	Estimated Direction	Planning Purpose
Dell PowerEdge R740XD Server	$1,900 - $3,200 depending on configuration/refurb tier	Core infrastructure host for Proxmox and VMs
Windows Server 2022 Licensing	$900 - $1,400 plus CAL planning	Domain, DHCP, DNS, NPS/RADIUS, infrastructure services
Windows 11 Pro Endpoints	$140 - $220 per license if needed by procurement path	Standardized desktop/laptop rollout
Microsoft 365 / Entra P1	Per-user subscription planning	Hybrid identity, SSO, MFA, Teams, OneDrive, admin visibility
Cisco / VPN / Support Licensing	Varies by firewall, support contract, and remote-access scope	Secure edge access, device supportability, remote operations

SECTION 4

Assets, Inventory & Procurement

After research established the direction, the next layer of the project focused on what would be tracked, procured, licensed, labeled, and maintained as the clinic moved into a more organized infrastructure model.

Asset tracking begins in Google Sheets stored in Google Drive as the early source of truth. Device name, model, quantity, serial information, ownership, location, status, and licensing notes are recorded there before later documentation is copied into broader operational repositories. The clinic's intent is to assign asset tags through a label machine and maintain manual, human-readable inventory discipline rather than depending entirely on automated scanning.

Selected Asset Direction

Core Server

Dell PowerEdge R740XD, ~64GB RAM, ~8TB storage, Proxmox host, Windows and Linux service VMs.

Endpoints

Dell OptiPlex fixed desktops and Dell Inspiron laptops for mobile and support needs.

Network Stack

Cisco ASA firewall, Cisco router, Cisco CBS 1300 switching, Ubiquiti APs, managed printer.

Recovery Hardware

Synology NAS and UPS integrated into the support and backup posture.

Inventory Fields Maintained

- Asset tag, hostname, device type, role, user assignment, location, model, serial number
- Support notes, warranty or sourcing references, license assignments, and procurement status
- Static IP notes for critical systems such as NAS, printer, firewall, switching, and management ports
- Relationship to documentation repositories, diagrams, and configuration records

SECTION 5

Documentation, Diagrams & Topologies

Good infrastructure becomes sustainable only when naming, diagrams, passwords, IP plans, management interfaces, and policies are recorded in reliable places. Summit Care's documentation model is built around a practical multi-location repository approach.

Documentation storage model: active documentation is maintained in a shared Office Drive / OneDrive space, copied into the office repository, and backed up to the Synology NAS so the project maintains a three-location documentation posture for resilience and access continuity.

Naming Standards

Category	Standard	Example
Desktop PCs	SCMC-PC-###	SCMC-PC-001
Routers	SCMC-RT-###	SCMC-RT-001
Switches	SCMC-SW-###	SCMC-SW-001
Access Points	SCMC-AP-###	SCMC-AP-001
Domain Controller VM	SCMC-DC-###	SCMC-DC-001
VPN VM	SCMC-VPN-##	SCMC-VPN-01
User Accounts	First initial + last name	sdaniel

Documentation Contents

- Asset inventory spreadsheets, device location records, and label references

- Hostname, IP address, VLAN, subnet, server role, and application role documentation
- Group policy summaries, firewall rule notes, VPN settings, DHCP reservations, and wireless standards
- SSH access standards, private key handling, password manager workflow, and certificate-backed RADIUS notes
- Diagram families for physical, logical, virtual, VPN, hybrid cloud, and management topology views

Management GUI Reference

Platform	Access Pattern	Purpose
Proxmox	Local management IP / secured admin path	Manage VMs, host resources, snapshots, and virtualization operations
Synology NAS	Local static IP plus QuickConnect/cloud visibility	Backups, shares, recovery files, documentation copies
UniFi / Network Manager	Cloud-connected or local controller visibility	AP management, wireless status, client visibility, support review
Printer GUI	Local static IP management page	Printer status, network settings, support tasks
iDRAC	Dedicated management subnet	Out-of-band server recovery and hardware management

SECTION 6

Infrastructure Deployment & Configurations

Deployment for Summit Care is based on golden baselines, repeatable build standards, and documented administration. The purpose is not only to stand the environment up once, but to make it supportable, auditable, and scalable.

Core Deployment Stack

- Proxmox is installed on the Dell PowerEdge R740XD as the type-1 hypervisor.
- Windows Server 2022 VM provides Active Directory, DNS, DHCP, and NPS/RADIUS.
- Additional Linux VPN VM supports software-based backup remote access.
- Test/sandbox VM supports validation before changes are pushed broadly.
- iDRAC provides out-of-band recovery and management if host-level issues occur.

Endpoint Deployment Method

Workstations are provisioned from USB-based Windows 11 media using Rufus and the Windows ISO. After installation, endpoints are renamed according to the SCMC standard, joined to Active Directory, brought under Group Policy, and loaded with the approved software stack and OneDrive / Teams / Microsoft sign-in posture expected in the clinic.

DHCP Failover & Addressing Logic

DHCP is centered on Windows Server for internal address management because it aligns directly with the domain environment, reservations, DNS updates, and user support workflow. The Cisco router also retains DHCP capability as part of the continuity model. In practice, that means the Windows server handles the primary office scopes while the Cisco router remains configured to support fallback or scoped continuity if the Windows DHCP role becomes unavailable. Critical infrastructure devices such as the firewall, router, switches, NAS, printer, and iDRAC remain on documented static or reserved addressing rather than depending on dynamic leases.

The design intentionally keeps Windows DHCP as the normal source of truth because it integrates better with Active Directory, DNS behavior, reservations, and day-to-day

support. The Cisco router is not presented as a replacement for the Windows role; it is a continuity measure that allows basic scoped address service to remain available if the Windows DHCP role is down. This helps reduce downtime risk while preserving static visibility for every critical management and infrastructure device.

Cisco Configuration Workflow

Cisco devices are administered over secured management paths using SSH rather than open, unauthenticated access. Private keys are created, stored securely, and referenced through the documented admin workflow. Running configurations are saved, exported, and retained in multiple protected locations so the team can rebuild or re-import configurations if hardware must be replaced. The documented workflow includes validating the running config, writing changes to startup config, exporting backup copies, and retaining those copies on the server, NAS, and cloud documentation locations.

VPN Configuration & Redundancy Model

Summit Care's remote-access design uses two different VPN paths. The Cisco ASA serves as the primary edge VPN because it sits directly in the firewall path and can enforce the same perimeter restrictions used elsewhere in the office security posture. A Linux-based software VPN virtual machine is also deployed within the Proxmox environment as a secondary remote-access path. That internal software VPN provides administrative recovery flexibility and preserves remote support access if the primary firewall-side VPN path must be serviced, changed, or temporarily taken offline.

This dual-VPN model is important because it supports both resilience and control. Staff and IT can rely on the ASA as the normal business-facing path, while the Linux-based VPN remains documented and available for backup continuity. The two paths are not meant to create uncontrolled duplication; they exist so remote access remains recoverable without improvising insecure access during an outage or maintenance event.

Automation & Scheduled Operations

PowerShell scripts are stored on the server's protected local volume and run through Task Scheduler to handle recurring support tasks such as backup routines, operational checks, log review triggers, and alert workflows. Linux-based support routines complement the

Windows side where needed, while SNMP monitoring provides threshold-based visibility into device health and service stability.

SECTION 7

Hybrid Cloud Infrastructure & Scalability

Summit Care's hybrid cloud approach is centered on Microsoft identity and service integration, not on moving every workload into the cloud. The local server environment remains the operational anchor while Entra and Microsoft 365 extend identity, password management, collaboration, licensing, and cloud-side user administration.

Hybrid Rollout Method

1. Microsoft Entra Connect is installed on the domain controller.
2. The local UPN / trust model is aligned to match the cloud identity format.
3. Synchronization is established so users and changes flow between local AD and cloud identity.
4. Password hash sync, password writeback, SSPR, and MFA are enabled as part of the hybrid identity posture.
5. Microsoft 365 apps, Teams, OneDrive, and related cloud services are tied into the same sign-in workflow.

Azure / Entra Features Actively Used

Feature	How It Is Used	Why It Matters
Entra Connect Sync	Connects local directory identities to cloud-side visibility	Lets IT manage one user story across local and Microsoft services
Password Hash Sync / Writeback	Allows password operations to remain consistent across environments	Reduces support friction and helps with account recovery
SSPR	Supports self-service recovery through the Microsoft side	Improves efficiency and user support flow

Feature	How It Is Used	Why It Matters
MFA	Applied organization-wide for Microsoft-connected access	Strengthens sign-in security and HIPAA-aware administrative posture
Teams / OneDrive / Microsoft Apps	Apps open under a cleaner SSO-driven user experience	Improves day-to-day productivity and reduces sign-in friction

A full Azure VM stack was considered as a future growth option, especially in the context of another office or expanded hybrid operations, but the current design intentionally stops short of that and focuses on hybrid identity, cloud app access, sync monitoring, and scalable Microsoft administration without overbuilding the first phase.

SECTION 8

Network Segmentation, VLANs & Subnetting

Segmentation is one of the main architectural controls that separates the upgraded Summit Care environment from a simpler flat office network. The design is function-based, not department-based, and is intended to enforce clearer traffic boundaries, easier ACL management, and lower lateral-movement risk.

Addressing Model

The office uses the private range **10.7.0.0/16** and breaks it into role-based /24 subnets for endpoints, internal servers, web-facing services, database/application data, guest Wi-Fi, cameras, VPN access, voice, general management, and out-of-band server management.

VLAN	Subnet	Role
10	10.7.10.0/24	User endpoints
20	10.7.20.0/24	Internal servers and Windows infrastructure services
30	10.7.30.0/24	Web/public services
40	10.7.40.0/24	Database/application data
50	10.7.50.0/24	Guest Wi-Fi
60	10.7.60.0/24	Cameras
70	10.7.70.0/24	VPN and remote access
80	10.7.80.0/24	Voice / Teams phone support traffic
90	10.7.90.0/24	Management interfaces
99	10.7.99.0/24	iDRAC / out-of-band management

Configuration & Control Model

- Cisco switching uses 802.1Q trunking to move multiple VLANs across shared links.
- Inter-VLAN routing is controlled through the routed/firewall layer rather than unrestricted open routing.
- Cisco ASA ACLs define which subnets may communicate and under what conditions.
- Windows firewall and host controls add another internal enforcement layer.
- Configuration changes are made over SSH, documented, saved, exported, and retained for recovery.

Endpoint Network

Primary office workstation and laptop footprint

Internal Services

AD, DNS, DHCP, NPS/RADIUS, supporting VMs

Protected Data & Management

Database, management, iDRAC, printer, NAS, monitoring

SECTION 9

Cybersecurity, Data Protection & HIPAA Compliance

Security at Summit Care is layered into the design rather than added after the fact. The architecture combines segmentation, identity-backed wireless access, role-based controls, Microsoft MFA, logging, DLP, firewalling, secure administration, SNMP monitoring, and recovery-aware documentation practices.

Network Hardening

Internal, guest, public, database, management, and remote-access paths are separated. pfSense-style proxy filtering and Cisco firewall controls reduce exposure around edge and exposed services.

DLP & Data Handling

Email filters, file-transfer restrictions, and policy-aligned handling of regulated content help reduce accidental or unauthorized movement of patient and office data.

Monitoring & Alerts

Event Viewer review, SNMPv3 monitoring, scheduled PowerShell checks, firewall logs, and email alerts to IT or stakeholders create a more operationally useful security posture.

Secure Access Practices

- WPA2-Enterprise wireless tied to Windows NPS/RADIUS with certificate-backed access where supported.
- Organization-wide MFA across Microsoft-connected services and management portals.
- SSH private key or certificate-based administration for Cisco and Linux systems.
- VPN access through Cisco firewall VPN plus Linux-based backup VPN path.
- Role-based access through AD groups, Entra visibility, and controlled app permissions.

SSH administration is treated as a controlled workflow rather than a convenience feature. Private keys or certificate-backed credentials are generated for approved administrators, stored in protected locations, referenced through the password manager and documentation process, and used only through known support paths. The Cisco ASA and Linux-based VPN model also supports this broader security design: the ASA remains the primary remote-access service at the edge, while the Linux-based software VPN provides a secondary path for continuity if primary access requires failover.

HIPAA-Aware Fit

Software and cloud-connected services were reviewed before adoption to confirm they better supported healthcare operations, privacy-aware administration, security visibility, and the regulated handling expectations relevant to Summit Care's environment.

SECTION 10

Disaster Recovery & Backup Plan

Summit Care's recovery design follows a practical 3-2-1 backup model supported by documentation, exported configurations, UPS-backed continuity, and rebuild baselines.

3-2-1 Recovery Layers

- Primary on-prem copy retained on the local server environment
- Secondary local copy retained on a Synology NAS with static IP and QuickConnect visibility
- Offsite / cloud-connected copy retained through OneDrive and related managed storage paths

Backup Execution Model

PowerShell scripts and scheduled tasks run daily at approximately 11:50 PM to back up the server's C drive, supporting system data, and the Practice Perfect database. This schedule is intended to reduce manual backup dependence and create a repeatable nightly protection cycle.

Recovery Readiness Measures

Recovery Control	How It Supports Recovery
UPS Protection	Protects the server during short power interruptions and unstable power events
Static Verizon Public IP	Keeps remote access and predictable service connectivity more stable
Config Exports	Cisco and Ubiquiti configs retained across server, NAS, and cloud locations

Recovery Control	How It Supports Recovery
Documentation Copies	Allows rebuild and validation even if one repository becomes unavailable
SNMP Alerts	Email alerts notify IT and stakeholders when monitored thresholds or failures occur

Because DHCP continuity, baseline documentation, exported configurations, and image-based rebuild methods are all part of the design, the clinic can recover more predictably if systems fail, need replacement, or require clean reprovisioning from known-good standards.

SECTION 11

Conclusion

This Summit Care Medical Clinic master report captures a complete network modernization workflow from planning to recovery readiness. Presented through a fictionalized healthcare organization, it documents how a small office environment can be redesigned into a more structured enterprise-style production model through stronger documentation, standardized deployments, hybrid identity, segmented networking, monitored security controls, and layered recovery design.

As a portfolio document, this report is intended to show design reasoning, implementation logic, and infrastructure leadership in a form that is detailed enough to demonstrate real technical depth while remaining safe to share for interviews, professional review, and resume support. It reflects not just the final architecture, but also the workflow discipline required to take a loosely managed environment and convert it into a supportable, scalable, and security-conscious infrastructure blueprint.

Deliverable summary: one protected, presentation-ready blueprint that combines company goals, technical research, procurement direction, documentation standards, deployment methodology, hybrid cloud integration, segmentation design, cybersecurity controls, and disaster recovery planning into a single long-form report.

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